

Sun Tracking Energy Harvester

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I. Introduction

With global energy demand at an all-time high, and the impacts of climate change becoming increasingly severe day-by-day, the need for sustainable energy systems has never been greater. Modern strategies are needed, utilising long-lasting and low environmental impact technologies that maximise the use of sustainable energy production and promote environmental recovery. Around the world, nations are accelerating their transition to renewable energy in pursuit of a net-zero carbon future by 2050.

This project focusses on sun-tracking energy systems; technologies that are designed to follow the movement of the sun to maximise energy collection throughout the day. These systems enhance efficiency by adjusting to changing light conditions which in turn reduces unnecessary power consumption during sub-optimal periods, extends the lifespan of solar harvesting systems to make them more tolerant to harsh environments, and minimising land usage areas in solar farms while also lowering noise and visual disturbance, while protecting local wildlife and nearby residents.

An in-depth case study of the Ivanpah solar plant in the Mojave Desert, California, was used as a baseline for this project. An arid desert environment with minimal rainfall is an ideal condition for a solar plant; however, problems arise quite quickly with this location. The remote location of the plant means less disruption to human activities, but harder to access for maintenance engineers. Environmental challenges also arise from dust storms and thermal stresses, of which must be considered for any alternative ideas. Furthermore, endangered species inhabit the area along with rare desert plants. The biggest flaw of the Ivanpah is due to it needing fossil fuels to power the main tower for solar harvesting, making it not truly sustainable by definition.

The Ivanpah plant covers around 14km² (3500 acres) of land, yet only produces 400MW of power, equivalent to 300MWh per \$1Million invested. The plant was built to promote sustainable energy production through large-scale solar plants; however, as mentioned previously, many challenges came with it. There is reduced efficiency due to dust storms and high maintenance demands, along with high costs of deployment and resource-intensive manufacturing processes. There are ongoing debates over sustainability versus environmental disruption, which we aim to minimise. The Ivanpah uses natural gas to power the start-up mechanism, hence contributing to emissions and going directly against its primary purpose of promoting sustainable energy production.

Wind loading must also be taken into account, as this was identified as the largest concern and commonly overlooked. Wind is the leading cause of damage to panels, inhibiting its use and causing costly repairs, overall lowering the lifespan of the panels. Thermal stresses must also be looked at due to the -- location of the solar farm, the Mojave Desert, having concerning high peak temperatures partnered with low nighttime temperatures – typical of a desert climate.

These two factors will become key parts in later simulations of our system, ensuring that our solution meets necessary criteria.

The Mojave Desert, albeit a desert, is home to thriving flora and fauna. Namely the desert tortoise that inhabits this area, has been affected by the Ivanpah plant, displacing them from their habitat, as the plant inhibits their survival. Birds have also been observed to have severe burn marks, and in the worst case, have died from the intense concentrated solar rays produced from the mirrors used in the Ivanpah plant.

II. Concept Design Process

The concept design phase started off with understanding the scope of our project and working from there. The idea of tracking the sun must be included in every concept, however, any additional ideas were welcomed and considered. This early phase of the design process allowed for a wide variety of ideas to flow forward. The two best concepts will be discussed and then weighted against each other and against a standard stationary photovoltaic panel, to ensure a baseline was provided.

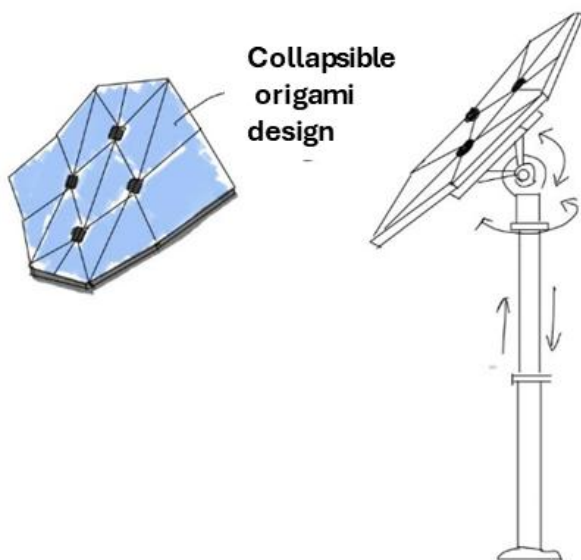


Figure 1.0

Folding Origami-Style Solar Panel with Dual Axis Movement

The first concept curated was a collapsible origami design in Figure 1.0, in which the panel was sought to close during the night, and during adverse weather conditions. This was intended to reduce the maintenance required from cleaning the panel, typically after rain or desert sandstorms. A dual-axis movement was added to ensure the panel can track the sun's azimuth efficiently, increasing energy harvesting compared to typical stationary photovoltaic panels.

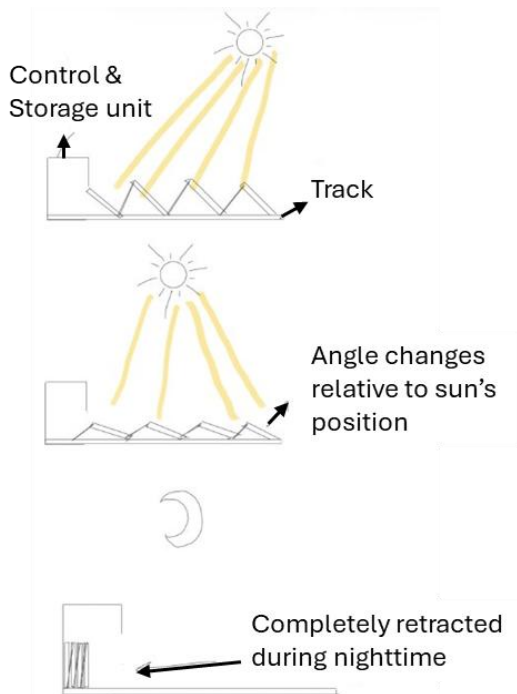


Figure 1.1

Concept drawing of retractable panel

Figure 1.1 shows initial designs for a closing panel, with a mechanism that is linked by rails, folding in at night and during harsh weather. There is housing for the panels, where maintenance can be done if the panel has damage, needs to be cleaned, or otherwise. The panels will extend out during the day depending on the sun's position, holding the angle at which energy harvesting is at maximum, before changing again as the sun moves. The initial thought for this design was to not have it move on a dual axis, compared to the likes of Figure 1.0, and further work was done on this concept in the form of basic 3D CAD modelling. This was done in order to fully understand the mechanisms that would be involved with a retracting panel.

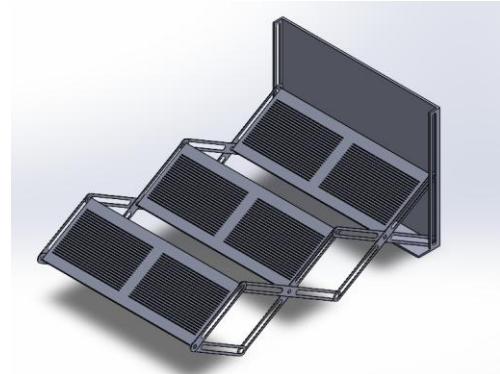


Figure 1.2

3D Model of retractable panel design

It was noticed during the modelling process for this design highlighted in Figure 1.2 that a vertical rail could be used instead of a horizontal rail. This was thought to be less complex, but after further thought, would still use the same process of extension as a horizontal rail, with the only difference coming from weight saving from a smaller rail. There would also be significant stresses on the backboard and vertical rail to support the weight of the panels and their frames. A key difference between concepts is the fact that Figure 1.1 and Figure 1.2 cannot move effectively to track the sun's azimuth, while Figure 1.0 can. This would ultimately lead to higher efficiency and better energy harvesting, which is heavily considered in the next section when weighing up the two concepts.

With it being deemed too difficult to move forward adding a dual-axis mechanism to the second concept design, the origami design was taken forward for my careful consideration.

III. Frame Design Process

An initial frame design was developed with the objective of minimising the environmental impact of the implementation of solar panels to the desert environment. A large portion of this impact is developed from the disturbance of desert ecosystems, including the destruction of habitats, wildlife and vegetation, which is often a consequence of extensive land clearing and the

use of concrete foundations for attaching the panels to a stable base.

To minimise these impacts, a tripod configuration was selected. This consists of hollow steel pipes welded together to form a stable support system. This design provides an attachment point for the motor and panel assemblies whilst reducing the need for large scale ground alteration.

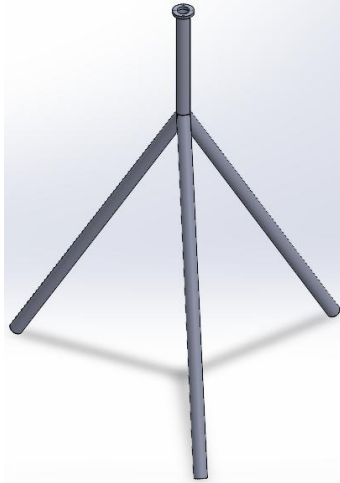


Figure 2.0

Tripod frame design to hold origami panel

Figure 2.0 shows the tripod frame, where it will be installed into the desert sand through the use of vibratory pile driving. This technique utilises controlled vibrations that generate high frequency vertical oscillations, therefore reducing the sand resistance by temporarily rearranging sand particles. As a result of this reduced friction, the frame is able to sink into the sand under its own weight or with a minimal external force. An earth pressure analysis was performed to estimate the embedment required for a tripod leg subjected to a 5kn lateral load applied 1m above the tripod, at the mounting point of the panel. Using the measured leg diameter of 115mm, a sand friction angle of 30° and a unit weight of approximately 650kg. The embedment depth was 1.34m with a factor of safety of 3.0.

This method offers several advantages such as reduced noise emissions, thereby minimising disturbance to the surrounding environment when compared to traditional impact hammers, and by not using concrete lowering overall carbon emissions for the whole implementation process. Concrete is an energy intensive material to produce, consuming a large amount of energy often from fossil fuel, thus, minimising the use of it was a key factor in designing a sturdy frame to hold the panels. While vibratory pile driving does generate vibrations, these are generally less disruptive to the ground structure than the repeated shocks produced by impact hammers. Furthermore, the technique enables the desert surface to remain largely intact,

preserving existing vegetation and wildlife, supporting the continued development for native species, most importantly the critically endangered Mojave Desert tortoise.

The continued development of the panel itself is discussed in latter parts of this report, and goes in detail about sizing, calculations and costs.

IV. Dual-Axis Mechanism

As mentioned briefly during the early design phase, the idea of a dual-axis mechanism to track the sun was proposed. This would ensure the panels angle to the sun is optimised at all stages of the day, leading to the highest output of energy. Although complex at first thought, two motors are to be used connected to gears that can spin the panel along two axes.

To do this we have made use of Nema motors mounted to the frame with bevel and pinion gears to achieve the movement of the panels in both the axis to achieve the tracking of the sun.

V. Materials Selection

For the selection of the materials to be used for the frame, ANSYS Granta EduPack was utilised. Initially a level 3 database was loaded and plotted onto a graph of Yield Strength (MPa) vs Price (USD/kg), shown in Figure 2.1.

For the next stage the decision was made to limit the materials to:

- Bulk materials and metals
- Fracture toughness $>25 \text{ MPa}\cdot\text{m}^{0.5}$
- Non-magnetic
- Good or excellent weldability
- Acceptable or excellent freshwater resistance
- Acceptable or excellent saltwater resistance
- Excellent UV radiation resistance
- Non-flammable
- Both recyclable and functionally recyclable

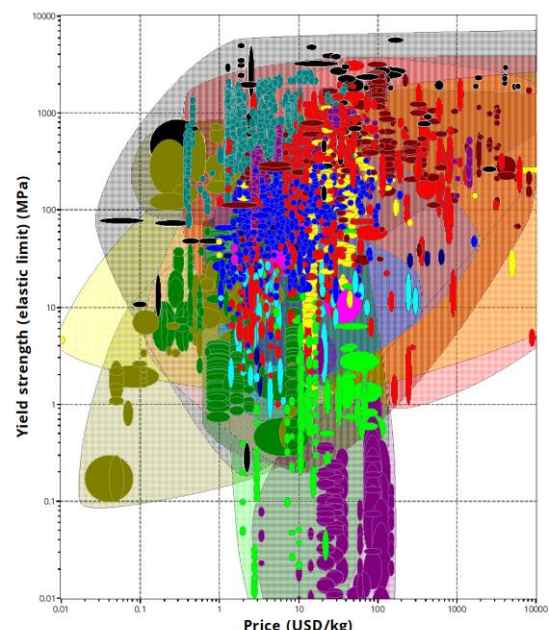


Figure 2.1 – Level 3 EduPack Database showing Yield Strength vs Price

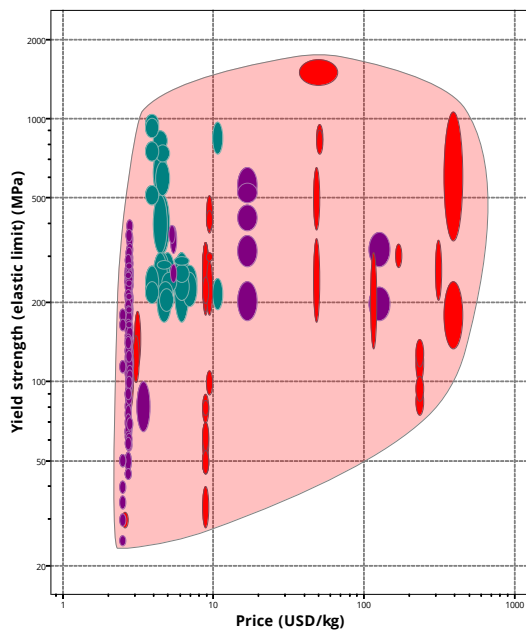


Figure 2.2 – Figure 2.1 including previously mentioned limits

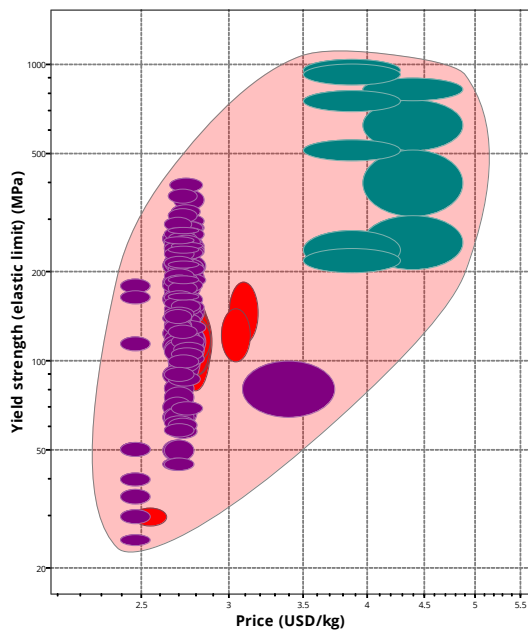


Figure 2.3 – Figure 2.2 including a limit of 5 USD/kg

To control overall costs, a threshold of 5 USD/kg was established. The shortlisted options were therefore evaluated by plotting yield strength (MPa) against embodied energy for typical grade production (MK/kg). Upon closer examination, stainless steel AISI 301 was selected as the final material choice. This grade not only exhibits the highest yield strength among the shortlisted options but also demonstrated

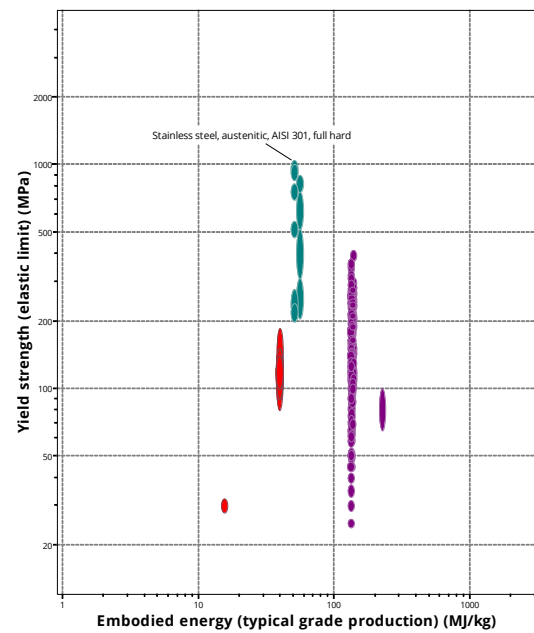


Figure 2.4 – Final suitable materials on a graph of yield strength vs embodied energy

comparatively low embodied energy during both primary production and post processing of the recycled material. Although stainless steel is generally more costly than aluminium, which is also recyclable, the reduced energy demands associated with its production and recycling were considered to outweigh the additional expense

VI. Project Goals for Calculations

The purpose of this project has been to improve the efficiency when compared to the Ivanpah plant, utilising tracking and folding systems, to ensure a more sustainable future for our planet. A number of targets were set by the team that acted as a guideline for the aforementioned design processes.

The Mojave Desert presents some of the harshest environmental conditions for any solar-tracking system. As detailed in the environmental analysis, the region experiences extreme daytime temperatures reaching up to 45°C, rapid thermal cycling between day and night, intense UV exposure, and frequent dust storms. Wind speeds can exceed 40 m/s, with design simulations reaching 100 km/h, and the variable, erosion-prone soil introduces further structural challenges. These desert conditions significantly reduce panel efficiency over time through soiling, thermal stress, and mechanical degradation, and they create some of the most demanding requirements for system safety, stability, and long-term

performance. Understanding these constraints was essential for shaping the performance targets for the tracking system.

The Ivanpah Solar Electric Generating System served as a critical benchmark because it demonstrates both the potential and shortcomings of large-scale solar energy installations in the Mojave Desert and other similar environments. Despite its intended output of over 1,000 GWh per year, Ivanpah produced only half of this during its first year, due to weather variability, soiling, and operational inefficiencies. Its reliance on natural gas for each morning startup also undermines its sustainability claims. Moreover, its vast land usage of 14.2 km² caused severe ecological disruption, including habitat loss and high wildlife mortality, such as bird deaths from concentrated solar flux. These drawbacks highlighted key areas where a more sustainable design must improve, reducing land usage, enhancing efficiency, lowering operational emissions, mitigating glare hazards, and ensuring robustness under environmental stresses.

The combined insights from Mojave environmental conditions and the shortcomings observed at Ivanpah directly informed the creation of the Measuring and Goals Table in the sub-systems calculations. The table defines minimum, medium, and maximum targets across technical, environmental, and economic dimensions, establishing a structured framework for evaluating the performance of the redesigned sun-tracking system. For example, goals relating to wind tolerance, thermal cycling resistance, and corrosion index stem from the known degradation mechanisms caused by desert climate and abrasive dust. Land-use efficiency, compactness ratio, and ecological disturbance targets were created in response to Ivanpah's oversized footprint and environmental impact. Energy-output goals aim to surpass Ivanpah's underperformance, particularly by improving tracking accuracy and reducing temperature-related power losses. Economic targets, including LCOE and cost-to-build, reflect the need for a more financially viable and resource-efficient system.

Ultimately, the goals table translates real-world challenges into quantifiable design criteria, allowing each subsystem's mechanical, structural, thermal, and electrical to be evaluated against measurable benchmarks. By establishing clear thresholds for durability, sustainability, efficiency, and cost, the table guides the engineering process toward a design that not only survives the Mojave Desert environment but outperforms legacy systems like Ivanpah. This structured approach ensures that the redesigned sun-tracking harvester addresses known environmental constraints while pursuing meaningful improvements in sustainability and operational performance.

Measuring and goals tables				
Parameter	Units	Minimum Goal	Medium Goal	Maximum Goal
Land Area Used	km ²	14.2 > X	10 > X	7.5 > X
Area per Mirror (Unit Area)	m ²	14 > X	12 > X	10 > X
Height of panel	m	2.2 > X	2 > X	1.8 > X
Width of panel	m	3.2 > X	3 > X	2.8 > X
Potential Life Cycle (NET)	years	12 < X	20 < X	30 < X
Maintenance & Downtime	% downtime / year	10 > X	5 > X	2 = X
Avoided CO ₂	Tons CO ₂ /Year	100'000 < X	400'000 < X	500,000 < X
CO ₂ Footprint	Tons CO ₂	43'000 > X	28'000 > X	20'000 > X
Cost to Build	£	1.76e9 > X	500e6 > X	300e6 > X
Cost to Run	£/year	26e6 > X	17.5e6 > X	12e6 > X
Net Output	kWh/year	117.5e6 < X	200e6 < X	300e6 < X
Energy per Land Area	kWh/m ² /year	8.25e6 < X	15e6 < X	25e6 < X
Compactness Ratio (retracted area vs operational area)	Ratio	1 > X	0.8 > X	0.7 = X
Ecological Disturbance Score	Qualitative	9 > X	6 > X	4 = X
Glare / Reflection Hazard	Lux / qualitative	9 > X	6 > X	4 = X
Dust Resistance / Soiling Rate	% output loss/day	1 > X	0.7 < X	0.5 = X
Wind Load Tolerance	km/h	27 < X	40.5 < X	100 < X
Heat resistance	°C	31.5 < X	54 < X	60 < X
Thermal Cycling Resistance	Cycles	5,000 < X	10,000 < X	20,000 < X
Corrosion Resistance Index	Rating (1–10)	6 < X	8 < X	9 = X
Tracking Efficiency Gain vs fixed	%	0 < X	20 < X	40 = X
Standby Energy Consumption	KWh/day	21'000 > X	10'000 > X	5'000 > X
End-of-Life Recyclability	% recyclable	60 < X	75 < X	90 = X
Embodied CO ₂ of Materials	ton CO ₂	165'000 < X	100'000 < X	50,000 < X
Fail-Safe Reaction Time (High Winds)	Seconds	40 < X	30 < X	20 < X
Ease of Assembly / Disassembly	Hours	2 < X	1 < X	0.75 (45 mins) < X
Levelized Cost of Energy (LCOE)	£ per kWh	0.5 < X	0.3 < X	0.15 = X
Return on Investment Time	Years	20 < X	15 < X	12 < X

Table 1.0 – Measuring and goal table

VII. Systems Calculations

The solar cells for this project were selected from MetSolar, a manufacturer known for producing high-quality monocrystalline photovoltaic cells and fully customisable solar modules. MetSolar offers a range of cell formats, including the 157.35 mm × 157.35 mm HTJ monocrystalline cells used in our calculations. These cells were chosen because they provide high efficiency (23.3%), which is essential for maximising power output within the compact footprint of the folding origami design. There are also customization options, allowing modules to be produced in non-standard shapes and layouts. This flexibility fits well with the project's geometric constraints, where triangular and square panels must integrate seamlessly into the folding mechanism.

- Cell dimensions: 157.35 mm × 157.35 mm.
- Cell area $A = 0.15735 \text{ m} \times 0.15735 \text{ m} \approx 0.02475 \text{ m}^2$.

Under standard test conditions (STC):

Solar irradiance, $G = 1000 \text{ W/m}^2$ and cell efficiency $\eta = 23.3\%$ (0.233).

Power per cell,

$$P_{cell} = \eta \cdot G \cdot A = 0.233 \times 1000 \times 0.02475 \approx 5.77 \text{ W} \approx 5.8 \text{ W}.$$

Three panel types used:

1. **5×5 square panel (750 mm × 750 mm):** This is a simple square panel built from a 5 by 5 grid of individual solar cells (25 cells total). This panel forms the centre of the assembly. There is a total of 4 square panel.
2. **Triangular panel 1 (base = 1500 mm, height = 750mm):** This triangular layout is formed by arranging cells in rows with 8 cells in the bottom row, 6 in the row above, then 4, then 2, then 1 at the top, with a total of 21 cells in total. This is the large triangular flap in the origami geometry which is responsible for folding and unfolding. There is a total of 4 triangular panel 1's.
3. **Triangular panel 2 (base, height = 750 mm):** A smaller triangular flap built from rows of 4, 3, 2 and 1 cells with a total of **10 cells**. There is a total of 12 triangular panel 2's.

These three tile shapes let the origami assembly combine square and triangular pieces so the folded structure packs tightly and the deployed shape gives a large collection area.

Panel geometry, volume and mass

1. Square panel area: $0.75 \times 0.75 = 0.5625 \text{ m}^2$.
2. Triangular panel 1 area: $\frac{1}{2} \times 1.5 \times 0.75 = 0.5625 \text{ m}^2$.

3. Triangular panel 2 area: $\frac{1}{2} \times 0.75 \times 0.75 = 0.28125 \text{ m}^2$.

Laminate thickness $t = 4.5 \text{ mm} = 0.0045 \text{ m}$ [ref].

Volumes:

- Square/ Triangular 1: $V = 0.5625 \times 0.0045 = 0.002531 \text{ m}^3$.
- Triangular 2: $V = 0.28125 \times 0.0045 = 0.0012656 \text{ m}^3$.

Material density (glass): 2500 kg/m^3 .

Assuming + ~10% for cells/encapsulant/backsheet.

Mass (Square/ Triangular 1):

$$m = V \times \rho \times 1.10 = 0.002531 \times 2500 \times 1.10 \approx 6.96 \text{ kg}.$$

Mass (Triangular 2):

$$m = 0.0012656 \times 2500 \times 1.10 \approx 3.48 \text{ kg}.$$

Origami assembly total mass: $(4 \times 6.96) + (4 \times 6.96) + (12 \times 3.48) \approx 98 \text{ kg} \pm 1 \text{ kg}$ (hinges, nuts, bolts etc.).

The tracking system is driven by an FY86EL400A two-phase stepper motor, selected due to its 4 A phase current, 0.97 Ω phase resistance and a holding torque of 8.5 Nm. Since stepper motors supply lower torque during motion, approximately 70–80% of the rated value is assumed usable, giving a dynamic torque of about 7 Nm. To ensure sufficient rotational authority against gravity and wind loading, this motor is paired with a 5:1 gearbox operating at 95% efficiency. The resulting output torque is calculated using $T_{output} = T_{motor} \times G \times \eta$, giving $T_{output} = 7 \times 5 \times 0.95 = 33.25 \text{ Nm}$, which is adequate for driving the dual-axis mechanism.

Energy modelling begins with the performance of the monocrystalline HTJ solar cells used in the panel arrangements. Each cell has an area of $A = 0.15735^2 = 0.02475 \text{ m}^2$ and an efficiency of 23.3%, allowing the power output under standard irradiance (1000 W/m^2) to be expressed as $P_{cell} = \eta G A$. Substituting values gives $P_{cell} = 0.233 \times 1000 \times 0.02475 \approx 5.77 \text{ W}$, rounded to 5.8 W per cell. Summing the outputs of the three panel groups (5×5, 8–6–4–2–1, and 4–3–2–1) gives a total of approximately 1.7 kWh per origami assembly. The tracking motors consume significantly less energy. Motor power is expressed as $P = I^2 R \times \text{phases}$. For the selected motor: $P = 4^2 \times 0.97 \times 2 \approx 31 \text{ W}$ per motor, and with four motors the total becomes 124 W. Daily operation includes opening, closing and tracking movements, amounting to 74.4 Wh/day. Net daily energy production is then determined by subtracting motor consumption from the gross solar output. In winter (7 hours of usable daylight), each assembly produces $1.7 \times 7 = 11.9 \text{ kWh/day}$; in summer (11 hours), output reaches

18.7 kWh/day. After subtracting the 0.0744 kWh/day required for tracking, the seasonal net values become 11.825 kWh/day and 18.625 kWh/day respectively. Averaging these gives a representative daily output of 15.225 kWh/day = 0.015225 MWh/day per assembly. To match the daily generation of the Ivanpah plant (2841 MWh/day), approximately $2841/0.015225 \approx 186,600$ assemblies would be required. Finally, land-use implications are evaluated by multiplying the footprint of one assembly (3.1 m² including clearance) by the number of required units. This gives a total land requirement of $A = 186,600 \times 3.1 \approx 578,460 \text{ m}^2 \approx 0.58 \text{ km}^2$. When compared to the 12 km² occupied by Ivanpah, the proposed system would reduce land usage by roughly 95%. This demonstrates that the origami-based modular design provides good spatial efficiency while maintaining equivalent energy output.

VIII. Validation of structures and life span

The structural behaviour of the sun-tracking panel system was evaluated under the key environmental loads present in the Mojave Desert, where high wind speeds, extreme temperature cycles and panel self-weight govern design performance (2) (3) (4) (5). Mechanical loading, thermal effects and combined structural response were assessed to ensure the frame, pole and actuator mechanisms can safely operate across all tracking positions (2).

This section will include load calculations and potential simulation-based analysis of:

- Normal Load cases (Gravity forces and CG changes)
- Wind loading cases (2) (7)
- Corrosion/Abrasion Degradation cases (4) (6) (8)
- Thermal Expansion and Cycling cases (3) (9) (10)
- Thermal effects on efficiency (9) (10)
- Torque required for moving parts

All cases mentioned here have either been simulated to present contour plots showcasing hotspots and points of interest (POI'S) or done on a handmade calculator as some parameters consistently change.

Other considerations will be discussed at the end of this section as some cases outside the listed ones above are too inconsistent to calculate however, they will still need to be discussed concerns and counter precautions against these risks. (2) (3)

Normal Load cases (Gravity forces and CG changes)

The simplest case for a sun tracking panel is the normal load cases. This case suggests that the only acting force on the panel is gravity and the weight/height from the ground is the only considered acting force on the system. However, systems like this aren't static and move/tilt (as shown in figure X) which in turn changes its centre of gravity (CG) which in turn affects the acting moment arm and there for force on the panel. (2) (3) (4) (5)

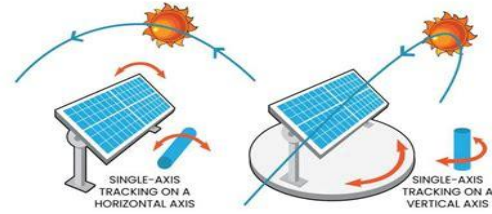


Figure 3.0
Dual axis system and Planes of movement/rotation

We can first state the simplest model calculation for our weight force is using Newtons laws of motions,

$$F = ma = g * m = W \text{ (Eqn 6.1)}$$

When the panel tilts, it creates a bending moment against the horizontal tilt axis when the force isn't acting in line with the CG. Torque due to gravity can be estimated as,

$$M_{Tilt} = W * d_{CG} \text{ (Eqn 6.2)}$$

W is the weight of the panel measured Weight of the panel measured in Newtons, and the d_{CG} is the horizontal off set of CG from the pivot measured as

$$d_{CG} = L_{offset} * \sin(\theta) \text{ (Eqn 6.3)}$$

Where L_{Offset} is the distance where the force is acting based on the offset and θ is the angle of tilt for the panel.

This typically occurs at intermediate angles such as 45°, when reaching tilts of 90° and 0°, the torque becomes extremely negligible. Also consider this can be acting in two directions as this panel is dual axis so this calculation will need to be done twice, once for the longitudinal way, and another for the lateral way.

In a typical configuration, the panel is mounted on a 1.5 m horizontal support arm connected to a vertical pole. The panel's weight of approximately 490 N produces a cantilever bending moment of around 735 Nm at the arm's base, along with a sheer force of 490 N. This load also applies an axial

compressive force to the pole and an equivalent bending moment due to the arm offset. If two symmetrical arms are used, each supports roughly half the load, reducing forces and moments per arm. In all cases, these loads must be balanced through static equilibrium to determine appropriate reactions at the support, actuators and foundation [1–4].

Wind loading cases

In evaluating wind loading, standard sea-level air properties were assumed, taking air density as $\rho = 1.225 \text{ kg/m}^3$. Following practices used at the Ivanpah Solar Farm, a design wind speed of 40 m/s was selected to provide a suitable factor of safety for extreme gust conditions in the Mojave Desert [1,5]. The simplest wind-loading case treats the panel as a flat surface exposed to uniform flow; however, a sun-tracking system does not remain fixed and continually changes orientation as it follows the sun. As the panel tilts and rotates, its projected area and aerodynamic behaviour vary, altering both the magnitude and direction of the wind forces acting upon it. In the worst-case scenario, the panel is oriented broadside to the wind at 90° , resulting in the maximum dynamic pressure and bending moment on the structure. These orientation-dependent effects must be considered to ensure the tracker maintains structural stability and safety across all operating positions. Angles. (3)

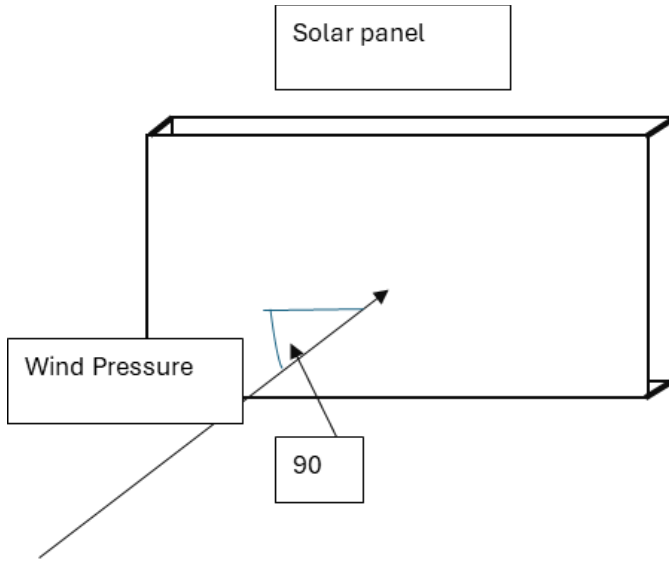


Figure 3.1

Diagram showcasing wind direction and AOA

$$q \text{ (Dynamic Pressure)} = \frac{1}{2} \rho V^2 \text{ (Eqn 6.4)}$$

This pressure corresponds to a flat plate fully exposed to wind at 90° . According to ASCE 7-16, ground-mounted PV panels are treated as either open-roof structures or freestanding signs depending on tilt angle, requiring the use of a pressure coefficient C_p to account for windward pressure and leeward suction. For a flat plate in high winds, C_p typically ranges from 1.2 to 1.5; therefore, a value of approximately 1.3 is adopted as a reasonable estimate for a dual-axis tracker under worst-case loading. Because the panel can rotate to any angle, the critical condition occurs when it is fully vertical and facing the wind. For a representative panel area of around 2.2 m by 3.2 m, this orientation produces the maximum projected area and the highest resulting wind force on the structure.

$$A_{proj} = \sin(\theta) * A \text{ (Eqn 6.5)}$$

A 40 m/s desert gust can generate wind forces of approximately 5 kN on a full-size panel, equivalent to the load of a 500 kg mass acting on the structure. As the panel tilts away from the wind, its projected area decreases and the applied force reduces accordingly, with loading roughly halved at a 45° orientation. In a horizontal stow position, the exposed area becomes minimal, resulting in a significant reduction in wind load. This highlights the importance of stowing panels flat during severe wind conditions to protect both the tracker and supporting structure [3,5].

$$F_{w,panel} = q * C_p * A_{proj} \text{ (Eqn 6.5)}$$

The supporting structure, such as the pole, also experiences wind drag. A typical round pole may see a wind force during strong winds. Although this is much smaller than the force on the panel itself, it still contributes to the overall load and creates bending in the pole. These combined forces must be considered when designing and sizing the structure.

$$F_{pole} = \frac{1}{2} \rho V^2 C_D A_{pole} \text{ (Eqn 6.6)}$$

Wind pushing on the panel creates a strong bending effect at the base of the structure and applies torque to the tracker's rotation axes. The support system must be capable of resisting this overturning effect, so it does not tip or deform. In addition, the pivot joints and actuators need to be strong enough to handle the twisting force caused by the wind, trying to rotate the panel.

$$M_{base} = F_{w,panel} * H_{full} \text{ (Eqn 6.7)}$$

The main support pole carries the entire tracker and is designed as a circular hollow steel section to efficiently resist bending, with wind loading acting as the dominant structural demand. The panel's self-weight is comparatively minor, so

the pole's section modulus and stiffness are selected primarily to withstand wind-induced bending moments at the base.

A final consideration is the potential uplift generated when wind acts on the underside of the panel. At certain angles of attack, pressure differences across the surface can produce an upward force like the lift effect observed on aircraft wings. Although the panel is not designed as an aero foil, this uplift can still pose a risk by increasing loads on hinges and mounts or, in extreme cases, contributing to overturning forces on the support structure. Ensuring the design can withstand such uplift scenarios is therefore essential for system safety [1,3].

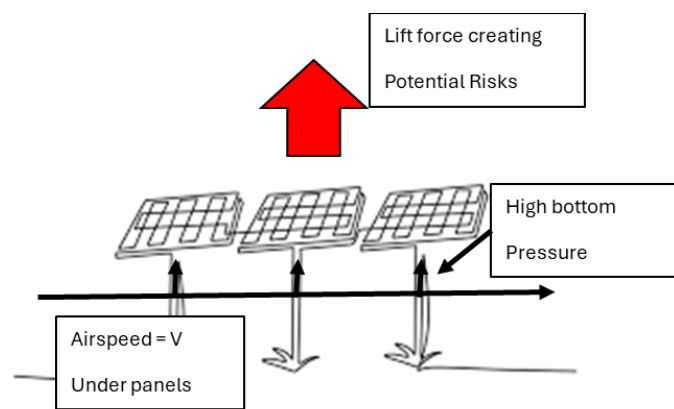


Figure 3.2

Figure demonstrating risk of potential hazard due to Lift Force

$$L = C_L * \frac{\rho}{2} * V^2 * A \text{ (Eqn 6.8)}$$

Where C_L (The coefficient of lift) in worst case scenarios can be calculated as 0.8 in high-speed winds when a flat surface is tilted between 10 – 15 Degrees. These calculations have been done based on naca foils which are made to create lift however our panel isn't, so it's a safe assumption to make as our panel will create more drag than lift.

Corrosion/Abrasion Degradation cases

Although the Mojave Desert is an arid environment with very low humidity, corrosion remains a notable concern for solar tracking systems. According to the corrosion and degradation table, oxidation is rated as a moderate risk due to the persistent presence of fine dust that can abrade protective coatings and expose bare metal surfaces. Occasional morning condensation and natural salts within the soil can further contribute to

localized corrosion, particularly at joints, fasteners and surfaces close to the ground [1,3].



Figure 3.3

Show casing piping in desert facing server degradation and corrosion

UV exposure in the Mojave Desert is classified as *high*, with intense sunlight throughout the year contributing to material fatigue and surface degradation, especially in polymers, cable sheathing and protective plastics [1,3]. Prolonged UV radiation can cause embrittlement, cracking and color loss, undermining the effectiveness of protective layers. Combined with thermal cycling daily swings of up to 35-40°C and seasonal swings of up to 55-60°C materials experience repeated expansion and contraction, which can break down adhesives, seals and coatings over time [1,2].

Materials used in the Mojave Desert are assessed for corrosion based on key environmental stressors such as oxidation, UV exposure, dust abrasion and salt or particulate intrusion. Corrosion risk is generally moderate, as the dry climate reduces moisture-driven rusting, but constant sand and dust erosion can remove protective coatings and expose bare metal surfaces. UV degradation is a significant concern, causing polymers and coatings to break down more rapidly under continuous sunlight, while dust and salt intrusion increases wear on joints, bearings and actuators. A corrosion resistance index, typically rated from 1 to 10, is used to indicate how well a material withstands these combined effects. Values around 6 represent minimum acceptable performance, with higher ratings of 8 to 9 indicating materials suitable for reliable long-term operation in harsh desert conditions [6,7].

Soil corrosive	Corrosion index C	Corrosion rate/(mm/a)
Weaker	0-0.2	0.1-0.2
Weak	0.2-0.4	0.2-0.3
Medium	0.4-0.6	0.3-0.6
Strong	0.6-0.8	0.6-1
Stronger	0.8-1	≥ 1

Figure 3.4

Show the Corrosion index used for rating materials and their suitability in harsh environments

Thermal Expansion and Cycling cases

The Mojave Desert undergoes extreme temperature fluctuations, with seasonal temperatures ranging from about -5 °C in winter to nearly 50 °C in summer. Daily swings are also large, typically from around 38 °C during the day to near 0 °C at night. These variations create significant thermal expansion and contraction in solar tracking structures. Solar modules can heat well above ambient conditions, often reaching 70-80 °C under direct sunlight, while exposed components may cycle between hot daytime temperatures and near-freezing nighttime values [1,3]. Overall, a temperature difference of around 55-60 K is a realistic design basis for assessing thermal stresses in the system.

Materials expand at different rates under temperature changes, described by their coefficient of thermal expansion (CTE). Carbon steel typically has a CTE of about 12×10^{-6} per °C, while aluminium expands almost twice as much at around 23×10^{-6} per °C. Fiber-reinforced composites show more complex behaviour, with glass-fibre materials generally ranging from 8×10^{-6} to 10×10^{-6} per °C and carbon-fibre composites as low as 0.2×10^{-6} per °C along the fibre direction. Because composites are anisotropic, their expansion varies by orientation, which must be considered when assessing thermal stresses in structural components [6,7].

Change in length due to heat can be measured as,

$$\Delta L = \alpha * L * \Delta T \text{ (Eqn 6.9)}$$

Where, ΔL , is the Change in length, α , is the Coefficient of thermal expansion, L is the length. And ΔT is the change in temperature from peak to peak (Night to Day). Again, we consider WCS (Worst Case Scenarios) where the Mojave desert experienced a peak temperature of 50°C to a drop of -5 °C, giving us $\Delta T = 55^\circ\text{C}$

When a material is restrained from expanding or contracting, thermal stresses develop because the imposed constraint

prevents natural movement. A fully fixed steel component subjected to typical Mojave temperature swings could experience stresses of around 139 MPa, more than half the yield strength of common structural steels. To avoid such high internal stresses, structures are generally designed with features like expansion joints or slight movement at connections, ensuring that temperature-induced deformation can occur without causing damage.

$$\sigma_{th} = E * \alpha * \Delta T \text{ (Eqn 6.10)}$$

Under the same temperature conditions, an aluminium component would experience a lower thermal stress of around 89 MPa. Although aluminium expands more than steel, its lower stiffness reduces the resulting stress. Even so, this value still represents roughly 30 percent of the yield strength of common aluminium alloys, reinforcing the need to account for thermal effects so components can safely accommodate expansion and contraction without risk of damage.

Thermal effects on efficiency

Solar panels are typically rated under “standard test conditions” which assume a cell temperature of around 25 °C. In real-world installations, especially under strong sunlight or in hot weather, cell (or module) temperatures can rise well above that reference. When that happens, the electrical performance of the panel usually degrades.

This occurs because the semiconductor inside the PV cells is temperature sensitive: as temperature increases, the material properties change in ways that reduce the voltage the cell can produce and thus lower its maximum power output. At the same time, heat also affects the supporting structure, mounting, and long-term durability of the installation. Thus, both the electrical performance and the structural design of a solar array need to account for temperature. [6,7]

PV panels lose efficiency as temperature rises, characterized by a temperature coefficient typically between -0.3 and -0.5% per °C above 25 °C. This means that at operating temperatures of 45-65 °C, common under strong sunlight, panels may lose 10-15% or more of their rated output. The reduction occurs because increasing temperature lowers the open-circuit voltage significantly, while the short-circuit current increases only slightly, causing overall power to decline. Cooler conditions therefore improve performance, provided irradiance remains high. To estimate real operating temperatures, PV analysis tools use the Nominal Operating Cell Temperature (NOCT), which provides a simplified method for predicting module temperature under actual environmental conditions. A common simplified formula is:

$$T_{\text{cell}} = T_{\text{ambient}} + \frac{(T_{\text{NOCT}} - 20)}{800} E_{\text{actual}} \quad (\text{Eqn 6.11})$$

Where,

- T_{cell} is the estimated cell (module) temperature in °C,
- T_{ambient} is the ambient air temperature in °C,
- T_{NOCT} is the NOCT rating for the module (°C) i.e. the module temperature under the standard reference conditions of NOCT,
- E_{actual} is the actual solar irradiance on the module (in W / m²).

More detailed calculators (or models used in PV software) sometimes base the estimate on a thermal-balance approach. They consider energy absorbed from sunlight, convective and radiative losses. In that case one solves (or approximately) something like:

$$P_{\text{abs}} = P_{\text{conv}} + P_{\text{rad}} \quad (\text{Eqn 6.12})$$

With,

$$P_{\text{abs}} = (1 - \eta) E_{\text{actual}} A \quad (\text{Eqn 6.13. a})$$

$$P_{\text{conv}} = h A (T_{\text{cell}} - T_{\text{ambient}}) \quad (\text{Eqn 6.13. b})$$

$$P_{\text{rad}} = \varepsilon \sigma A (T_{\text{cell}}^4 - T_{\text{ambient}}^4) \quad (\text{Eqn 6.13. c})$$

where A is the area of the module or cell, η is module efficiency, h is the convective heat transfer coefficient, ε is emissivity of the surface, and σ is the constant Stefan-Boltzmann. Solving such equations (often iteratively) gives a more physically realistic estimate of module temperature under various irradiance, wind, and mounting conditions.

Once the module temperature is estimated, the calculator then uses the module's "temperature coefficients" to correct the expected electrical output. For example, the open-circuit voltage V_{oc} (or the maximum power) can be adjusted by:

$$V_{oc}(T) = V_{oc,STC} + TC_{V_{oc}} \times (T - 25^\circ\text{C}) \quad (\text{Eqn 6.14})$$

$$P_{\text{max}}(T) = P_{\text{max,STC}} + TC_P \times (T - 25^\circ\text{C}) \quad (\text{Eqn 6.15})$$

Temperature coefficients such as $TC_{V_{oc}}$ and TC_P quantify how voltage and power change per degree Celsius, referenced to standard test conditions at 25 °C. By combining these coefficients with a cell-temperature calculator, engineers can estimate realistic panel performance under actual weather conditions rather than ideal laboratory values. This enables thermal analysis of the panel to predict operating temperatures, associated power losses and the impact of heat on structural components. It also helps assess whether frames, mounts, adhesives and electrical interconnects can tolerate thermal stress without deformation or failure. Overall, this approach provides a clearer understanding of real-world performance,

highlights areas most affected by thermal loading and supports verification that the panel design remains reliable under prolonged exposure to high temperatures.

Below, are the given materials and geometries for the simulation used in Ansys for thermal patterns in materials. These panels use the same materials as the ones made in the simulation. It has been done in this shape as it would give similar results while making the simulation more straight forward and there for more trustworthy.

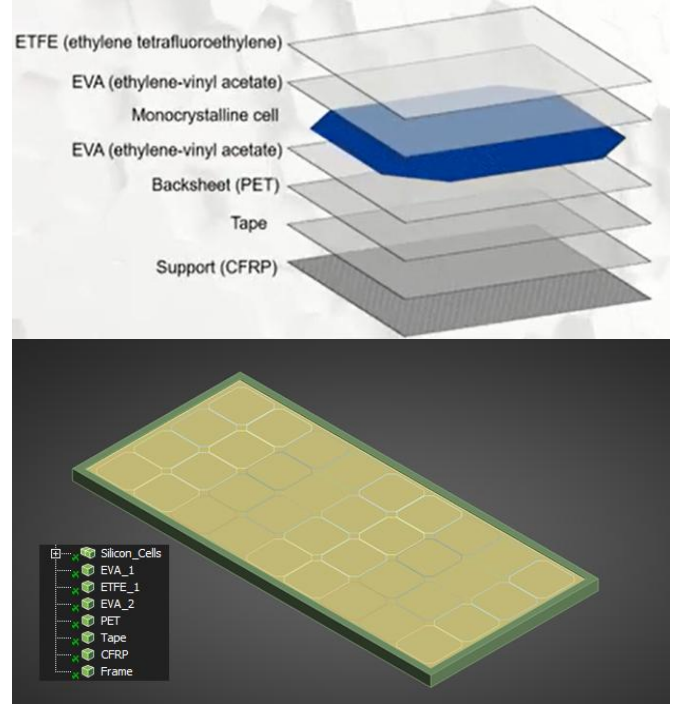


Figure 3.5
Materials patterns and panel geometry used in Ansys simulation

Below shows Figure 3.6, detailing the boundary condition setup from the given calculations done in the above section. These calculations are inputted from the excel calculator and ran based on WCS conditions in the Mojave Desert

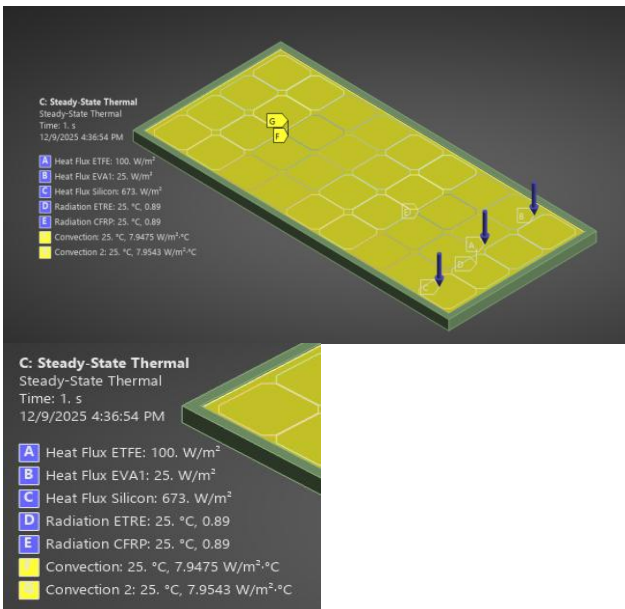


Figure 3.6
Boundary conditions for Thermal Simulation

Figure 3.7.a highlights distinct temperature gradients between individual cells and illustrates how heat moves across the silicon layer in Mojave Desert conditions. The edges stay colder because of increased convective heat loss, while hotspots form toward the centre where airflow is restricted and absorbed irradiance is highest. The pattern shows that uneven heating can happen even in a uniform material, which could hasten thermal fatigue, raise electrical resistance, and lower efficiency. [8–10]

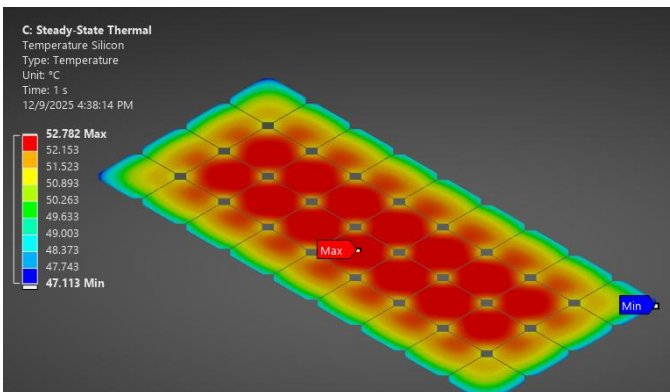


Figure 3.7.a
Thermal contour plot from the Mojave Desert simulations on just the silicon panels

Figure 3.7.b shows the rate at which thermal energy is escaping various silicon layer regions. Higher heat flux zones, which frequently match the hotspots in Figure 3.7.a, are

associated with regions with stronger irradiance exposure or larger temperature differences. Inefficient heat dissipation, which can trap heat and exacerbate localized thermal stress, is indicated by lower flux areas. This highlights the necessity of components and mounting arrangements that facilitate efficient cooling in high-load areas. [8–10]

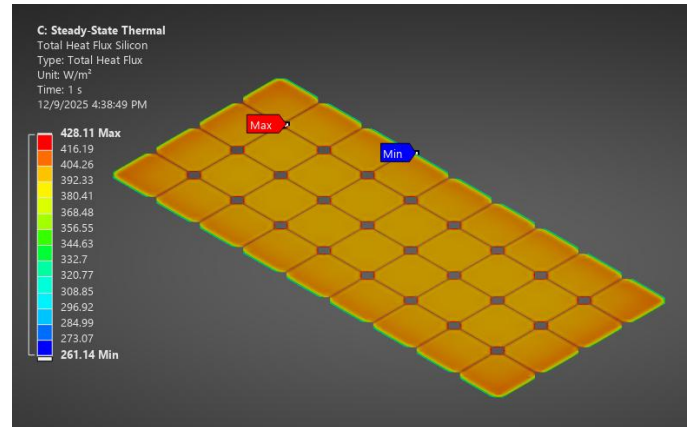


Figure 3.7.b
Heat flux emitted from the Mojave Desert simulations on just the silicon panels

In contrast to the bare silicon layer, heat spreads differently when the entire panel stack is considered Shown in Figure 3.7.c. Heat is redistributed by structural elements like encapsulants, backing layers, and the CFRP support, resulting in smoother but wider warm areas throughout the panel. This illustrates how the entire panel assembly raises the operating temperature by trapping and buffering heat. These findings are representative of actual Mojave conditions, where low wind and high irradiance can push panels into high temperature ranges that have an impact on long-term material durability and efficiency. [8–10]

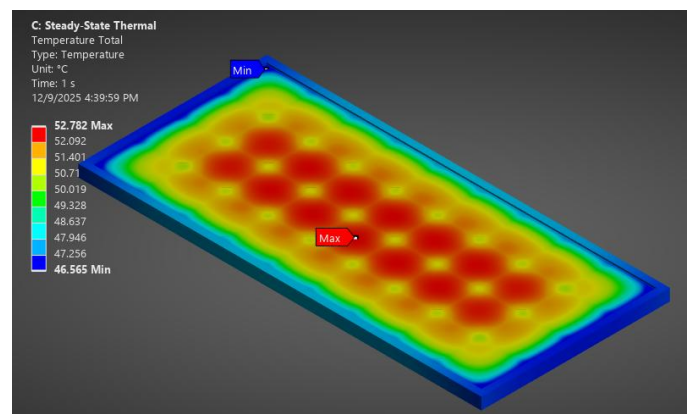


Figure 3.7.c
Thermal contour plot from the Mojave Desert simulations on the full panel

Torque required for moving parts

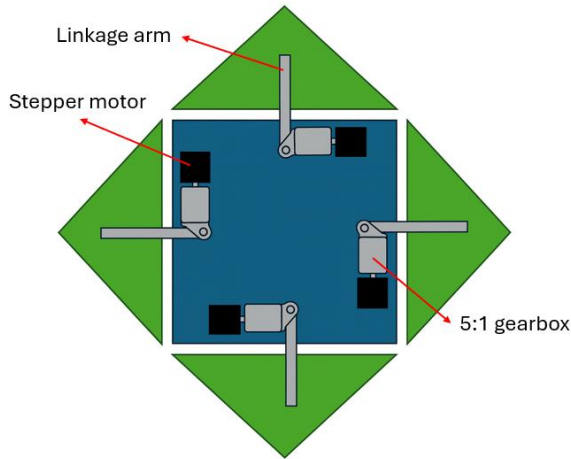


Figure 4.0
Folding mechanism for the panels



Figure 4.1
Panel and Frame assembly

Figure 4.0 and 4.1 show how each of the four folding triangular panels is actuated using a combination of a stepper motor, a gearbox, and a linkage arm. The blue square in the centre of Figure 4.0 represents the fixed base panel, which houses all the motors and supporting mechanisms. Each corner contains a stepper motor, which provides the rotational input needed to move the folding panels. Each folding panel is driven by a small stepper motor, which connects to a gearbox; the gearbox output drives a linkage arm that opens and closes the triangular flap. The gearbox is required because the triangular panel's centroid creates a lever arm that demands more torque than the bare stepper can reliably provide during motion and under external loads (for example wind). Placing the motors under the base keeps wiring and mechanisms protected while the gearboxes increase torque and give finer, safer control of the folding motion. [8–10]

Calculations

Geometry

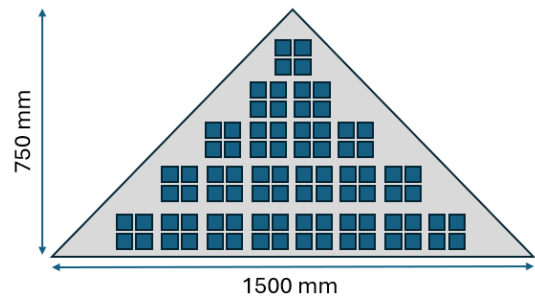


Figure 4.1.
Geometry of folding panels

- Panel mass $m = 6.96 \text{ kg}$
- $g = 9.81 \text{ m/s}^2$
- Weight, $W = mg = 6.96 \times 9.81 = 68.3 \text{ N}$
- Triangle base $b = 1.5 \text{ m}$.
- From area, $A = \frac{1}{2}bh$, height $h = 0.75 \text{ m}$.
- Centroid distance from base hinge $r = \frac{h}{3} = \frac{0.75}{3} = 0.25 \text{ m}$

Torque required at the hinge (panel horizontal)

$$T_{\text{required}} = W \cdot r = 68.3 \times 0.25 = 17.1 \text{ N/m}$$

Motor chosen:

Model	Step Angle (°)	Motor Length (mm)	Rated Current (A)	Holding Torque (N.m)	Phase Resistance (Ω)	Phase Inductance (mH)	Rotor Inertia (g.cm ²)	Lead Wires (NO.)	Motor Weight (Kg)
FY86ES350A	1.8	80	3.5	4.50	1.0	4.4	1500	4	2.00
FY86EM400A		94	4.0	6.00	0.8	3.5	2700	4	2.80
FY86EL400A		118	4.0	8.50	0.97	5.5	4100	4	3.80
FY86EC500A		150	5.0	12.0	1.20	6.0	6200	4	5.20

Table 2.0

Motor metrics

Usable dynamic torque of the FY86EL400A stepper is about **7N/m** (rated holding torque $\approx 8.5 \text{ N}\cdot\text{m}$; moving torque is lower).

Since $T_{dynamic} = 7\text{N/m} < T_{required} = 17.1\text{N/m}$, the motor by itself cannot supply the needed torque at the hinge.

Required torque with a safety factor

Using a $1.5\times$ safety factor:

$$T_{SF} = 1.5 \times 17.1 = 25.7 \text{ N/m}$$

Because the stepper alone cannot supply enough torque to lift the triangular panels, its output shaft is connected to a **5:1 gearbox**. The gearbox increases the available torque while reducing the rotational speed to a more controllable level.

Motor + 5:1 gearbox calculation

Motor usable torque $T_{motor} = 7\text{N/m}$, gearbox ratio $i = 5$, gearbox efficiency $\eta_g = 0.95$.

Output torque at the hinge (Transmitted to the linkage):

$$T_{out} = T_{motor} \times i \times \eta_g = 7 \times 5 \times 0.95 = 33.25 \text{ N/m}$$

Compare with requirement:

$$T_{out} = 33.25 \text{ N/m} > T_{SF} = 25.7 \text{ N/m}$$

So, the 5:1 gearbox gives a comfortable margin.

Checking the required motor shaft torque to achieve the hinge torque:

$$T_{motor, req} = \frac{T_{required}}{i \eta_g}$$

This is well below the motor's usable $7 \text{ N}\cdot\text{m}$, so the motor will not be overloaded.

Attached to each gearbox output is a linkage arm, which is the long grey bar extending toward the corresponding triangular panel as shown in Fig 4.1. When the motor rotates, the gearbox amplifies the torque and drives the linkage arm, causing the triangular panel to open or close. The orientation of each assembly is mirrored on all four sides so that each green triangular panel is actuated independently but follows the same mechanical principle. This arrangement keeps all moving components protected beneath the central base panel while allowing the external folding panels to deploy smoothly and symmetrically.

Additional considerations

Along with the problems listed above to validate that the product can with stand various harsh conditions in the Mojave Desert, there are also some other considerations which need to be talked about. They fall in this section as there either isn't enough valid data to reasonably calculate parameters for them, or the cases are too inconsistent to have a set of varied data to run on them. Even so, they still pose threat to disruptions or loss of life, so they need to be discussed.

These instances include:

- Seismic activity / Earthquakes
- Light glare
- Ground / Terrian conditions

Seismic activity / Earthquakes -

Since the Mojave Desert is in an active seismic zone, solar tracking systems are seriously at risk from earthquakes (). Shaking and vibration of the ground can cause trackers to misalign, joints to become loose, foundations to be damaged, and mechanical parts that depend on precise movement to be strained. Seismic loading creates an additional failure pathway that can shorten the lifespan and efficiency of structures that are already subjected to harsh desert conditions. Wider stances, flexible joints, reinforced or base-isolated foundations, and shock-absorbing dampers that permit controlled movement during seismic events can all help lower these risks. Early design and analysis that incorporates earthquake load cases is in line with sustainable design principles and contributes to long-term dependability, reduced maintenance requirements, and safer operation in the Mojave environment ().

Light glare -

According to the environmental parameters for the area, reflective glare is a significant problem in the Mojave Desert, where bright flashes can be seen up to 3 km away due to intense sunlight and reflective panel surfaces. This glare can interfere with wildlife, distract drivers and pilots, and endanger worker safety (). When tracking panels accidentally direct sunlight toward observers, this risk is increased. Designers can use textured surfaces, low-gloss or anti-reflective coatings, and safe stow angles to reduce glare. Additionally, glare-path modelling during design guarantees that reflections are diverted away from habitats, roads, and airplane routes ().

Ground / Terrian conditions –

The Mojave Desert's topography and ground conditions present several structural difficulties for solar-tracking devices. Due to mixtures of compacted ground and loose sand, the area has a moderate load-bearing capacity but highly variable soil stability. Additionally, there is a high potential for erosion due to continuous wind-driven movement. If foundations are not properly designed, these conditions may eventually cause them to shift, tilt, or deteriorate. Systems need ground anchors, deep or widened foundations, and erosion-resistant footing designs to prevent this. Protective skirts surrounding bases, grading, and soil compaction all aid in stabilizing structures. Long-term stability is ensured, and maintenance requirements are decreased by considering uneven terrain and soil variability. () ()

IX. Costing

A domestic supplier based in the United States was identified for the provision of the frame material, thereby reducing the transportation related emissions due to its proximity. Sandmeyer Steel was selected as it is capable of supplying the required grade of stainless steel at a cost of \$2.61 per kilogram. Given that a complete frame has an approximate mass of 99kg, the total cost was calculated to be £258.35.

The panel movement assembly incorporates the system's primary motion mechanism, which is composed of several mounting components and associated hardware. These elements are manufactured from the same material as the structural frame to simplify production and streamline the supply chain. The assembly includes mounting brackets and gears, collectively accounting for £333.47 of the cost. Additionally, the mechanism requires Nema 23 stepper motors priced at £21.82 each, as well as standardised mounting hardware such as nuts and bolts, costing £1.20 per pair. In total, the panel movement assembly incurs a cost of £387.60.

The PV panel assembly comprises 20 panels of varying sizes and geometries, designed to fold compactly to provide mechanical protection during stowage. The configuration includes four primary square panels, 4 large triangular panels and 12 smaller panels. Custom sized panels were sourced from the supplier MetSolar at a unit cost of £180 per m^2 . Based on the required surface area for all components, the total cost of the complete PV panel assembly is £1417.56.

The final component of the complete assembly is the panel folding mechanism. This mechanism incorporates four NEMA 34 stepper motors, each paired with a 5:1 NEMA planetary gearbox to reduce rotational speed and increase torque, thereby enabling the panels to fold effectively. The cost of

each motor- gearbox pair is £ 61.06. Additionally, the assembly requires panel housings as well as linkages and hinges to facilitate panel movement. The total cost for the entire folding mechanism amounts to £604.96.

This brings the total cost of all components of the sun tracking solar array to £2668.48.

X. Lifecycle Analysis

LCA Goal

The goal of this study is to look at how well the Sun Harvester solar panel system works for the environment over its entire life cycle. The main goal is to figure out which parts and steps have the biggest effect on carbon emissions, resource use, and toxicity. The study aims to identify environmental hotspots and suggest ways to improve design efficiency by looking at material choices, manufacturing processes, and operational needs. The assessment also aims to promote greater modularity, longer component life, and less waste at the end of life. The results will help develop plans that make the solar panel system more environmentally friendly, recyclable, and sustainable.

Boundaries of the system

The study has a cradle-to-grave scope (as shown in Figure 5.1), which means it looks at everything from getting the raw materials to making all the parts of the solar panels, shipping them, installing them, and running them with replacement cycles. It also includes processes for decommissioning and end-of-life, like recycling and throwing things away in a landfill. This makes sure that all of the environmental effects are fully assessed throughout the system's lifecycle.

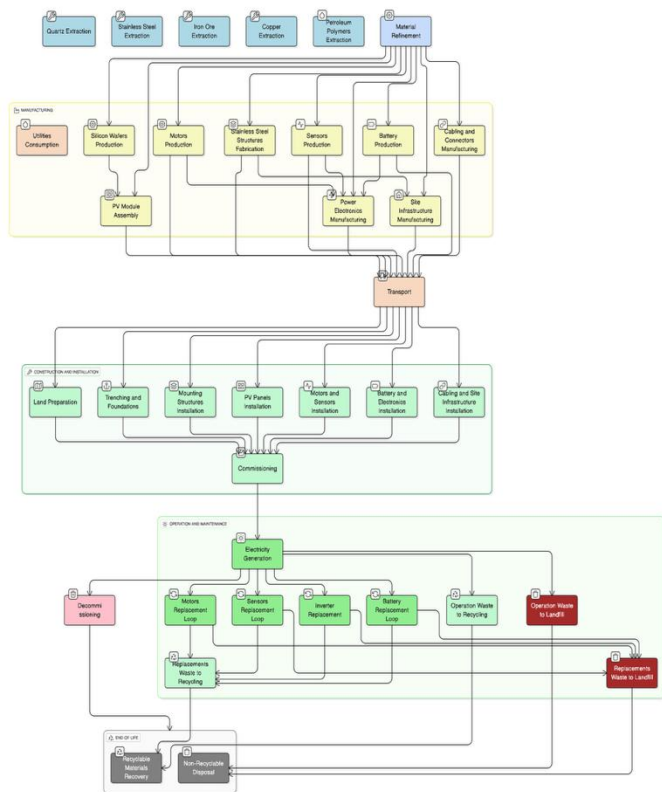


Figure 5.1
LCA cradle to grave scope

Functional Unit

The functional unit of this study is 1 kWh of AC electricity delivered to the grid by one PV tracking system over a 25-year lifetime. The complete system generates 138,878 kWh in 25 years, and all material, transport, and replacement flows are calculated for this lifetime output. Results are then divided by 138,878 kWh to express environmental impacts per 1 kWh, ensuring consistency, comparability, and alignment with ISO 14044 methodology.

Inventory Analysis

The Bill of Materials (BOM) made for the PV tracking system was used to make a list of all the parts. We used the technical data we had to figure out the mass and composition of each part. For some parts, like motors, gearboxes, sensors, and electronic units, where exact weights weren't publicly available, reasonable guesses were made based on engineering specifications and values found in the literature. Table 2.0 shows the final weights of all the parts that make up the PV tracking system.

Material group	Mass per set (kg)
Tempered glass	63.7
Stainless Steel AISI 301	8.8
Crystalline silicon	4.9
EVA	8.3
Copper	2
Stainless Steel AISI 302	86.91
Stainless Steel AISI 303	9.87
Stainless Steel AISI 304	1.89
Stainless Steel A4 316	0.248
4 gearboxes	10
6 motors	22.8
Cables	10
controller	1.5
sensor	0.1

Table 3.0 – Masses included in the assembly

The PV tracking system's complete material inventory was made by listing all the parts that would be needed over its 25-year lifespan. Using the Bill of Materials (BOM) and other software like EduPack Granta and the Ecoinvent database, we found out the mass and material makeup of each part, such as the PV module, structural elements, motors, gearboxes, cabling, junction boxes, sensors, and electronics.

It was necessary to calculate the lifetime material demand for some parts because they need to be replaced while they are in use. This was done by multiplying the mass of one unit by the number of replacements expected over 25 years. This made it possible to figure out how much material the system used overall, including parts that wear out, like motors, gearboxes, protection devices, and sensors.

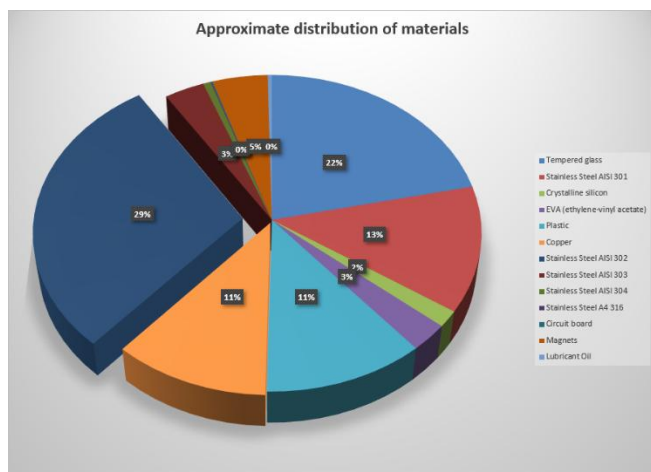


Figure 5.2 – Pi Chart showing material distribution

To connect the material to the functional unit, we divided the total amount of material used over the lifetime of the tracking system by the total amount of electricity it produced (138,878 kWh). This gave us material intensity values (kg/kWh and g/kWh), which show how much of each material is needed to make one kilowatt-hour of electricity. Fig 5.2 illustrates the overall material distribution of the system.

Impact Assessment:

Figure 5.3 shows the midpoint impact results for the PV tracking system. The bars show that the most important factors are freshwater ecotoxicity, terrestrial ecotoxicity, and human non-carcinogenic toxicity. This shows that the system uses a lot of materials, especially metals and electronic parts. Mineral resource scarcity is also a big problem, which shows that we rely on metals like copper, stainless steel, and permanent magnets.

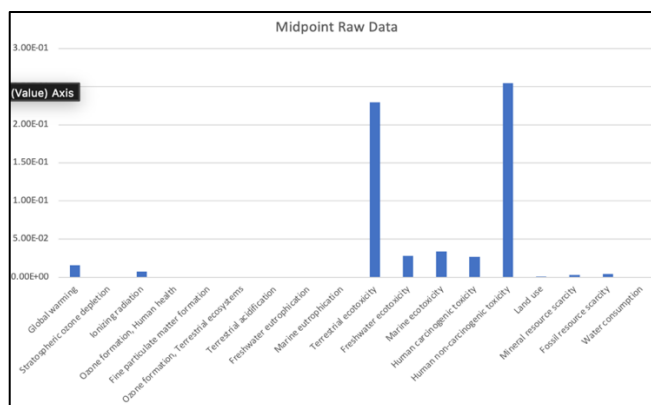


Figure 5.3
Midpoint impact

Figure 2X shows how these effects differ by material group, showing how different materials affect the main groups. Silicon wafers are mostly responsible for carbon emissions because making them takes a lot of energy. Polyurethane and PCB materials add smaller amounts to a number of different groups. Permanent magnets, on the other hand, have the biggest effect on a number of toxicity categories, especially terrestrial ecotoxicity, because of how rare-earth minerals are mined and how magnets are made.

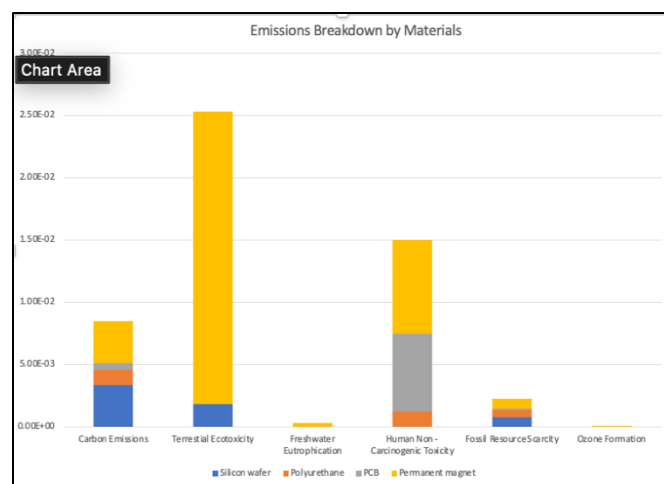


Figure 3X shows the main things that cause toxicity. It shows how stainless steel, copper ore, silicon wafers, and permanent magnets all add to ecotoxicity and human toxicity indicators. Stainless steel is shown to be the most toxic to freshwater ecosystems, which makes sense because processing alloys that contain chromium and nickel releases pollutants. The extraction of copper ore and permanent magnets is what makes humans toxic. This figure makes it very clear which materials are most to blame for the system's toxic burdens.

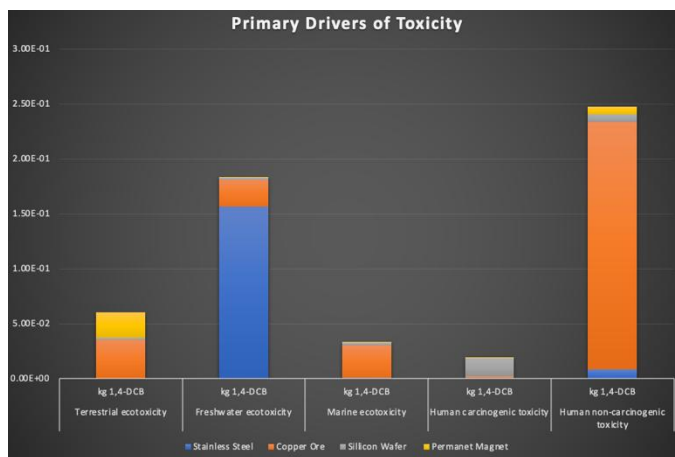
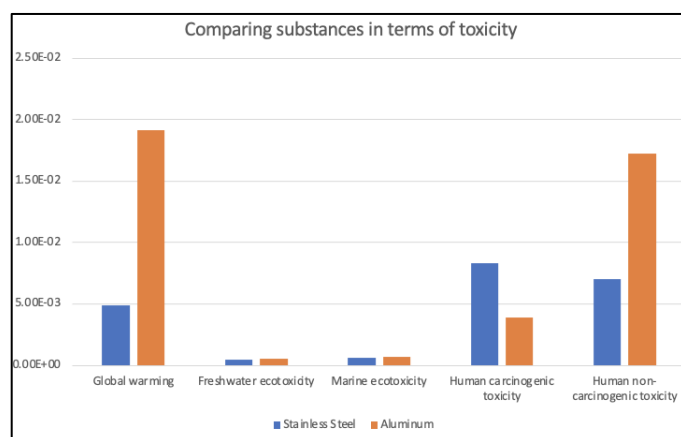


Figure 4X shows a direct comparison of stainless steel and aluminium in important toxicity categories. Aluminium always has higher impact values, especially for global warming, human non-carcinogenic toxicity, and ecotoxicity. The Hall–Héroult process used to smelt aluminium uses a lot of energy, and bauxite mining and red-mud waste cause problems for the environment.



XI. Conclusions and Future Work

The analysis conducted throughout this project indicates that the proposed sun-tracking energy harvester achieves several of the key performance targets established at the outset, particularly in areas related to land-use reduction, improved

energy capture efficiency, and structural resilience under typical Mojave Desert conditions. The dual axis tracking system demonstrates clear advantages over fixed photovoltaic panels, increasing total energy output and allowing the system to maintain optimal solar incidence throughout the day. The structural validation work confirms that the tripod foundation significantly reduces ecological disturbance compared with concrete-based alternatives, aligning with sustainability goals for minimal ground impact.

However, the system does not meet all targets. The bending stress produced under worst-case wind loading exceeded the material's yield strength (227 MPa vs. 200 MPa), indicating that reinforcement or material revision is required to prevent failure. Thermal stresses remain within allowable limits, but the thermal simulations highlight notable hotspots, showing that long-term temperature cycling may pose durability concerns. Certain environmental risks, such as seismic activity, ground instability, and glare reflection, also require additional engineering refinement to ensure compliance with safety and regulatory expectations.

Overall, the system successfully meets many sustainability, and performance aims but requires iterative development to fully satisfy the maximum goal criteria.

To enhance performance, reliability, and long-term sustainability, the following areas are recommended for future development:

Improve Structural Safety Under Extreme Winds

Wind loading remains the critical failure mode. Future work should:

- Increase section modulus of the support structure (larger diameter tube or thicker walls),
- Consider alternative materials with higher yield strength,
- Refine the stow-angle control strategy to reduce projected wind area automatically,
- Investigate aerodynamic shaping to reduce drag and uplift forces.

Expand Validation of Seismic, Ground, and Glare Responses

The three environmental risk categories require deeper analysis:

- **Seismic activity:** Perform modal analysis and soil–structure interaction studies to validate dynamic

response and recommend base-isolation or damping systems.

- **Ground/terrain effects:** Conduct geotechnical surveys to validate tripod penetration depth, soil compaction behavior, and long-term erosion effects.
- **Glare reflection:** Model glare trajectories across seasons and implement anti-reflective coatings or passive shading features to mitigate hazards to aircraft, wildlife, and nearby roads.

Explore Alternative Tracking Technologies

Current tracking uses sensor-based or algorithmic actuation. Future systems could incorporate:

- **Camera-based sun tracking** with computer vision for improved accuracy in variable conditions,
- **Edge-AI controllers** that optimize movement patterns to reduce energy consumption,
- **Hybrid predictive models** that blend astronomical tracking with real-time irradiance data.

These approaches may enhance tracking precision while reducing mechanical fatigue.

Material Innovation and Cost Reduction

As PV technology evolves, panel efficiency and material sustainability will continue to improve. This opens opportunities to:

- Adopt next-generation PV materials (e.g., perovskites, tandem cells) with improved efficiency and lower thermal sensitivity,
- Use lighter, stronger composite structures to reduce wind loading and material mass,
- Reduce manufacturing and lifecycle emissions by integrating recycled or low-embodied energy materials.

Refinement of Energy Modelling and LCA

Expanding energy modelling to include seasonal variation, long-term soiling behaviour, and degradation rates would enable more accurate lifetime predictions. A full cradle-to-grave lifecycle assessment (LCA) should also be conducted to quantify sustainability benefits over existing solar-thermal plants such as Ivanpah.

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XII. Bill of Materials

Part/Item Identification										Material and Mass				Cost Details (£)			Supply Chain Management			
Project Code	BoM Section	UID	Issue	Revision	Type	Part Description	Part Number	Full CAD Description	Quantity	Material	Density	Mass Each	Mass Total	Cost Each	Cost Total	Company	Supplier Part Number	Lead Time	Hyperlink	
STT2	00	0001	01	A	Assembly	Frame Assembly	STT2-00-0001-01	STT2-00-0001-01 Frame Assembly	1	Stainless Steel AISI 301	7880	98.986	98.986	£ 258.35	£ 258.35	Sandmeyer Steel	To Drawing	1 Week		
STT2	00	0002	01	A	Part	Tripod Member	STT2-00-0002-01	STT2-00-0002-01 Tripod Member	3	Stainless Steel AISI 301	7880	28.970	86.910	£ 75.61	£ 226.84	Sandmeyer Steel	To Drawing	1 Week		
STT2	00	0003	01	A	Part	Leg Mount Member	STT2-00-0003-01	STT2-00-0003-01 Leg Mount Member	1	Stainless Steel AISI 301	7880	9.870	9.870	£ 25.76	£ 25.76	Sandmeyer Steel	To Drawing	1 Week		
STT2	00	0004	01	A	Part	Flange	STT2-00-0004-01	STT2-00-0004-01 Flange	1	Stainless Steel AISI 301	7880	1.890	1.890	£ 4.93	£ 4.93	Sandmeyer Steel	To Drawing	1 Week		
STT2	00	183-9146	01	A	Standard Part	M12X50 Bolt	183-9146	M12X50 Bolt	4	Stainless Steel A4 316	7870	0.062	0.248	£ 1.58	£ 6.32	RS Components	183-9146	1 Day	RS PRO Stainless Steel Hex Bolt, M12 x 50mm RS	
STT2	00	530-078	01	A	Standard Part	M12 Nut	530-078	M12 Nut	4	Steel Zinc Plated	7850	0.017	0.068	£ 0.13	£ 0.52	RS Components	530-078	1 Day	RS PRO, Bright Zinc Plated Steel Hex Nut, DIN 934, M12 RS	
STT2	01	0007	01	A	Assembly	Movement Assembly	STT2-01-0007-01	STT2-01-0007-01 Movement Assembly	1	Mixed	N/A	129.757	133.592	£ 387.60	£ 387.60	N/A	To Drawing	1 Week		
STT2	01	23H545-42045	01	A	Standard Part	Nema 23 Stepper Motor	23H545-42045	23H545-42045 Nema 23 Stepper Motor	2	Mixed	4902	1.800	3.600	£ 21.82	£ 43.64	SKYSmotor	23H545-42045	5 Days	https://www.skysmotor.co.uk/ipod8/2363-Nema-23-Stepper-Motor-23H545-42045-3789-18-deg-38m-4-Wires.html	
STT2	01	0002	01	A	Part	Rotation Pinion Gear	STT2-01-0002-01	STT2-01-0002-01 Rotation Pinion Gear	1	Stainless Steel AISI 301	7880	0.968	0.968	£ 2.53	£ 2.53	Sandmeyer Steel	To Drawing	1 Week		
STT2	01	0003	01	A	Part	Rotation Bevel Gear	STT2-01-0003-01	STT2-01-0003-01 Rotation Bevel Gear	1	Stainless Steel AISI 301	7880	31.699	31.699	£ 82.73	£ 82.73	Sandmeyer Steel	To Drawing	1 Week		
STT2	01	0004	01	A	Part	X Plate	STT2-01-0004-01	STT2-01-0004-01 X Plate	1	Stainless Steel AISI 301	7880	32.699	32.699	£ 85.34	£ 85.34	Sandmeyer Steel	To Drawing	1 Week		
STT2	01	0004	01	A	Part	Motor Bracket	STT2-01-0004-01	STT2-01-0004-01 Motor Bracket	1	Stainless Steel AISI 301	7880	4.059	4.059	£ 10.59	£ 10.59	Sandmeyer Steel	To Drawing	1 Week		
STT2	01	287-4201	01	A	Part	M24 Nut	287-4201	M24 Nut	12	Steel Grade 8.8	7850	0.185	2.220	£ 0.71	£ 8.52	RS Components	287-4201	1 Day	https://uk.rs-online.com/web/p/nuts-mts/2874201	
STT2	01	0005	01	A	Part	Mounting Plate	STT2-01-0005-01	STT2-01-0005-01 Mounting Plate	1	Stainless Steel AISI 301	7880	20.875	20.875	£ 54.48	£ 54.48	Sandmeyer Steel	To Drawing	1 Week		
STT2	01	0006	01	A	Part	Support Stand	STT2-01-0006-01	STT2-01-0006-01 Support Stand	1	Stainless Steel AISI 301	7880	37.472	37.472	£ 97.80	£ 97.80	Sandmeyer Steel	To Drawing	1 Week		
STT3	01	530-163	01	A	Standard Part	M12 Bolt	530-163	M12 Bolt	4	Steel Grade 8.8	7850	0.047	0.188	£ 0.49	£ 1.96	RS Components	530-163	1 Day	https://uk.rs-online.com/web/p/hex-bolts/0530163	
STT2	02	0003	01	A	Assembly	Panel Assembly	STT2-02-0003-01	STT2-02-0003-01 Panel Assembly	1	Monocrystalline Bi-Facial Panel	2500	17.400	17.400	£ 1,417.56	£ 1,417.56	MetSolar	To Drawing	2 Weeks		
STT2	02	0001	01	A	Standard Part	Square PV Panel	STT2-02-0001-01	STT2-02-0001-01 Square PV Panel	4	Monocrystalline Bi-Facial Panel	2500	6.960	27.840	£ 101.25	£ 405.00	MetSolar	To Drawing	2 Weeks		
STT2	02	0004	01	A	Standard Part	Large Triangle PV Panel	STT2-02-0004-01	STT2-02-0004-01 Large Triangle PV Panel	4	Monocrystalline Bi-Facial Panel	2500	6.960	27.840	£ 101.25	£ 405.00	MetSolar	To Drawing			